BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

CE-403: Database Fundamentals (PostgreSQL)

Elchin Salahzada

Lecture 3

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

SQL Fundamentals II

- Cascades
- Constrains
- Type Conversion
- Joins: Inner Join
- Joins: Outer Join (Left, Right, Full)
- Self Join
- Subqueries
- Common Table Expressions (CTEs)
- Set Operations (UNION, INTERSECT, EXCEPT)
- Lab Activity

Learning objectives

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- Create tables using cascade functionality
- Restrict column using constraints
- Convert datatypes of the columns
- Combine data from multiple tables using INNER JOIN
- Retrieve matched and unmatched records using LEFT, RIGHT, and FULL OUTER JOIN
- Apply self joins to compare rows within the same table
- Write subqueries to filter or compute values based on results of another query
- Use Common Table Expressions (WITH clause) to simplify complex queries
- Combine result sets from multiple queries using UNION, INTERSECT, and EXCEPT

Cascades

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- ON DELETE CASCADE - if the primary data is deleted then foreign data also will be removed automatically
- ON DELETE RESTRICT - It does not allows you to delete the primary row from foreign table
- ON DELETE SET NULL - allows you to replace foreign data with null values when primary is deleted
- ON DELETE SET (default value) - instead of null you replace with specific number

Constraints

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

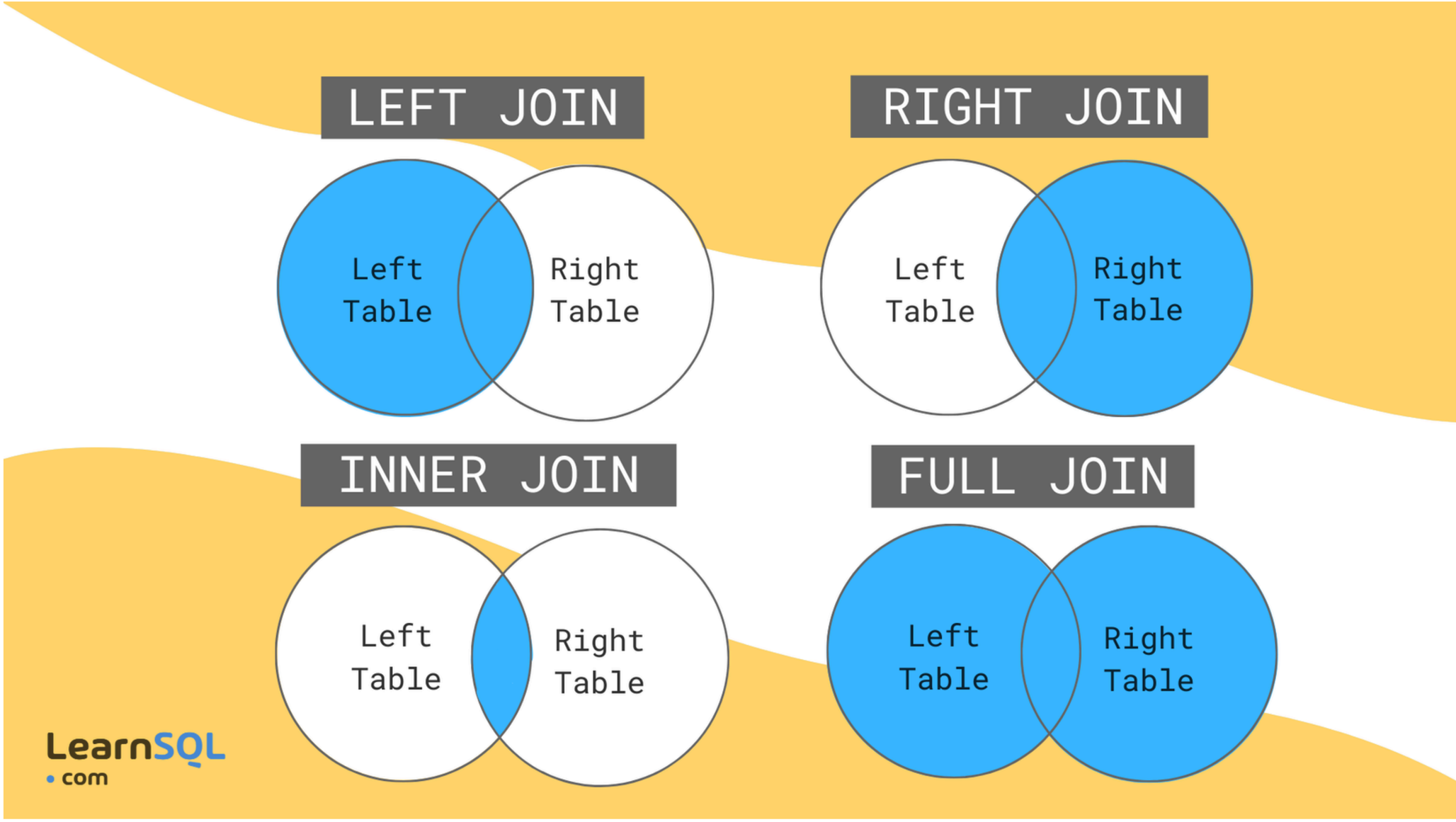
- PRIMARY KEY - defines unique value for the table
- FOREIGN KEY - refers to corresponding primary key in another table
- NOT NULL - specified column requires value
- UNIQUE - specified column cannot be duplicated
- CHECK - requires condition to be true while inserting data
- DEFAULT - gives default value for corresponding column while inserting data

Type Conversion

- CAST (column_name as data_type) - changes data type of given column
- column_name :: datatype - (recommended) changes data type of given column

Joins

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL



Joins: Inner Join

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- INNER JOIN returns rows that have matching values in both tables
  - SELECT columns
  - FROM table1
  - INNER JOIN table2
  - ON table1.column = table2.column;
- Example:
  - SELECT e.name, d.dept_name
  - FROM employees e
  - INNER JOIN departments d
  - ON e.dept_id = d.dept_id;
- Rows with no match in either table are excluded from results
- JOIN alone (without specifying type) defaults to INNER JOIN in PostgreSQL
- Table aliases (e, d) make queries shorter and more readable

Joins: Outer Join (Left, Right)

- Outer joins return matched rows plus unmatched rows from one or both tables
- LEFT JOIN – all rows from the left table, matched rows from the right (NULL if no match)
    - SELECT e.name, d.dept_name
    - FROM employees e
    - LEFT JOIN departments d
    - ON e.dept_id = d.dept_id;
- RIGHT JOIN – all rows from the right table, matched rows from the left (NULL if no match)
    - SELECT e.name, d.dept_name
    - FROM employees e
    - RIGHT JOIN departments d
    - ON e.dept_id = d.dept_id;

Joins: Outer Join (Full), Cross

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- FULL JOIN – all rows from both tables, with NULLs where there's no match
  - SELECT e.name, d.dept_name
  - FROM employees e
  - FULL JOIN departments d
  - ON e.dept_id = d.dept_id;
- Use outer joins to find unmatched records (e.g., employees with no department, or departments with no employees)
- CROSS JOIN returns the Cartesian product of two tables — every row from table1 combined with every row from table2
- No ON condition is used (no matching logic)
  - SELECT columns
  - FROM table1
  - CROSS JOIN table2;

Self join

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

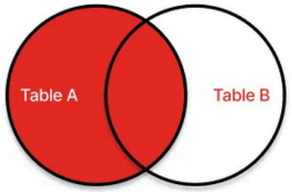
- A self join joins a table to itself — treated as two separate tables using aliases
- Useful for comparing rows within the same table (e.g., hierarchical relationships)
    - SELECT a.column, b.column
    - FROM table_name a
    - JOIN table_name b
    - ON a.common_column = b.common_column;
- Example: Find employees and their managers (both stored in the same employees table)
    - SELECT e.name AS employee, m.name AS manager
    - FROM employees e
    - JOIN employees m
    - ON e.manager_id = m.emp_id;
- Common use cases: organizational hierarchies, comparing records (e.g., employees with the same department or salary)

Joins

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

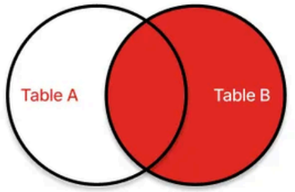
SQL JOINs Visualized with Venn Diagrams

LEFT JOIN



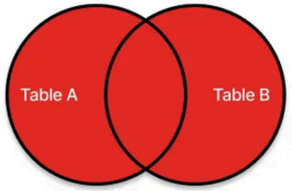

```
SELECT * FROM A LEFT JOIN B ON A.key = B.key;
```

RIGHT JOIN



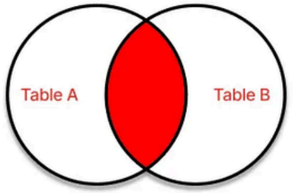

```
SELECT * FROM A RIGHT JOIN B ON A.key = B.key;
```

FULL OUTER JOIN



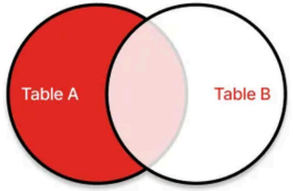

```
SELECT * FROM A FULL OUTER JOIN B ON A.key = B.key;
```

INNER JOIN



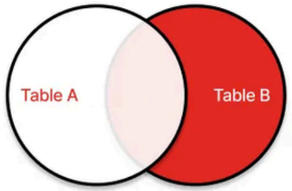

```
SELECT * FROM A INNER JOIN B ON A.key = B.key;
```

LEFT JOIN excluding INNER JOIN



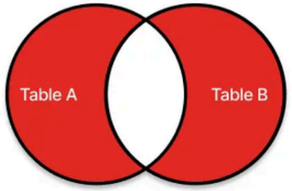

```
SELECT * FROM A LEFT JOIN B ON A.key = B.key
WHERE B.key IS NULL;
```

RIGHT JOIN excluding INNER JOIN



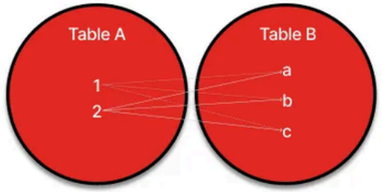

```
SELECT * FROM A RIGHT JOIN B ON A.key = B.key
WHERE A.key IS NULL;
```

FULL OUTER JOIN excluding INNER JOIN




```
SELECT * FROM A FULL OUTER JOIN B ON A.key = B.key
WHERE A.key IS NULL OR B.key IS NULL;
```

CROSS JOIN




```
SELECT * FROM A CROSS JOIN B;
```

Subqueries

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- A subquery is a query nested inside another query, enclosed in parentheses
- Can be used in SELECT, FROM, or WHERE clauses
- Single-row subquery – returns one value
    - SELECT name, salary
    - FROM employees
    - WHERE salary > (SELECT AVG(salary) FROM employees);
- Multi-row subquery – returns multiple values, used with IN
    - SELECT name FROM employees
    - WHERE dept_id IN (SELECT dept_id FROM departments);
- Correlated subquery – references the outer query, executes once per row
    - SELECT name, salary
    - FROM employees e
    - WHERE salary > (SELECT AVG(salary) FROM employees WHERE dept_id = e.dept_id);
- Subqueries can also appear in FROM (treated as a temporary table)

Subqueries (ANY, ALL, EXISTS)

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- ANY – compares a value to any value returned by the subquery (condition true if it matches at least one)
  - SELECT name, salary
  - FROM employees
  - WHERE salary > ANY (SELECT salary FROM employees WHERE dept_id = 2);
- ALL – compares a value to all values returned by the subquery (condition true only if it holds for every row)
  - SELECT name, salary
  - FROM employees
  - WHERE salary > ALL (SELECT salary FROM employees WHERE dept_id = 2);
- EXISTS – checks whether the subquery returns any rows (TRUE/FALSE), often faster than IN for large datasets
  - SELECT name FROM employees e
  - WHERE EXISTS (SELECT 1 FROM departments d WHERE d.dept_id = e.dept_id);
- NOT EXISTS – checks whether the subquery returns no rows
- = ANY(...) is equivalent to IN (...); <> ALL(...) is equivalent to NOT IN (...)

Common Table Expressions (CTEs)

- A CTE is a temporary named result set, defined using the WITH clause, used within a single query
- Improves readability by breaking complex queries into logical steps
    - WITH cte_name AS (
    - SELECT column1, column2
    - FROM table_name
    - WHERE condition
    -)
    - SELECT * FROM cte_name;

CTE Example

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- Example: Find employees earning above their department's average salary
  - WITH dept_avg AS (
  - SELECT dept_id, AVG(salary) AS avg_salary
  - FROM employees
  - GROUP BY dept_id
  -)
  - SELECT e.name, e.salary, d.avg_salary
  - FROM employees e
  - JOIN dept_avg d ON e.dept_id = d.dept_id
  - WHERE e.salary > d.avg_salary;
- Recursive CTEs – a CTE that references itself, useful for hierarchical data (e.g., org charts)
- CTEs vs. Subqueries: CTEs are more readable and can be reused multiple times in the same query

Set Operations

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- Combine results of two or more SELECT queries — both queries must have the same number of columns with compatible data types
- UNION – combines results, removes duplicates
    - SELECT name FROM employees
    - UNION SELECT name FROM contractors;
- UNION ALL – combines results, keeps duplicates (faster than UNION)
    - SELECT name FROM employees
    - UNION ALLSELECT name FROM contractors;
- INTERSECT – returns only rows that appear in both queries
    - SELECT dept_id FROM employees
    - INTERSECT SELECT dept_id FROM departments;
- EXCEPT – returns rows from the first query that don't appear in the second
    - SELECT dept_id FROM employees
    - EXCEPT
    - SELECT dept_id FROM departments;

Lecture Summary

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- INNER JOIN returns matched rows; OUTER JOINs (LEFT, RIGHT, FULL) also include unmatched rows
- CROSS JOIN produces a Cartesian product of two tables
- Self Join compares rows within the same table using aliases
- Subqueries nest one query inside another — single-row, multi-row, or correlated
- ANY, ALL, EXISTS, NOT EXISTS extend subquery comparisons against multiple values
- CTEs (WITH clause) improve readability and support recursive queries
- Set operations (UNION, UNION ALL, INTERSECT, EXCEPT) combine results from multiple queries
- Constrains are used for defining columns properties
- Cascade defines key relations between tables
- Type conversion are defined mostly in where clause

Hands-on

BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

- Scenario: Extend the company_db database with joins, subqueries, and set operations
- Create a contractors table (similar structure to employees) and a manager_id column in employees using constraints and cascades and superkey.
- Write an INNER JOIN between employees and departments
- Write a LEFT JOIN to find departments with no employees
- Write a CROSS JOIN example (e.g., employees × departments)
- Write a self join to find employees and their managers
- Write a subquery to find employees earning above the overall average salary
- Write a subquery using IN to find employees in departments located in a specific city
- Write queries using ANY, ALL, and EXISTS
- Write a CTE to find employees earning above their department's average salary
- Write UNION, INTERSECT, and EXCEPT queries comparing employees and contractors names



BAKI ALİ NEFT MƏKTƏBİ
BAKU HIGHER OIL SCHOOL

Next Lecture

- Aggregate Functions & Grouping